

Pathogenesis

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1st Stage

LG 2

Introduction

- Most bacterial infections are acquired from an external source.
- However, some bacterial infections are caused by members of the normal flora and, as such, are not transmitted directly prior to the onset of infection.

Pathogenesis

- **Pathogenesis:** is the process by which an infection leads to disease.
- **Pathogenicity:** is the ability of a microbe to produce disease in a susceptible individual.

Pathogenesis

- **Pathogen:** A microorganism capable of causing disease.
- An organism may be described as a frank or true pathogen or as an opportunistic pathogen.

Pathogenesis

- **True pathogens:** are organisms recognized to cause disease in healthy immunocompetent individuals.
- Bacterial species such as *Yersinia pestis* and *Bacillus anthracis* are pathogenic in nearly all situations

Opportunistic pathogen

- **Opportunistic pathogen:** An agent capable of causing disease only when the host's resistance is impaired (when the patient is “immunocompromised”).

Carrier:

- **Carrier:** A person or animal with asymptomatic infection that can be transmitted to another susceptible person or animal.

Infection:

- **Infection:** Multiplication of an infectious agent within the body.
- Multiplication of the bacteria that are part of the normal microbiota of the gastrointestinal tract GIT, skin, and so on is generally not considered an infection; on the other hand, multiplication of pathogenic bacteria (eg, *Salmonella* species) even if the person is asymptomatic is deemed an infection.

Virulence

- **Virulence:** is a quantitative measure of pathogenicity and is measured by the number of organisms required to cause disease.
- The 50% lethal dose (LD50) is the number of organisms needed to kill half the hosts
- 50% infectious dose (ID50) is the number needed to cause infection in half the hosts.
- Organisms with a *lower* LD50 (or ID50) are said to be *more* virulent than those with a higher LD50 (or ID50) because fewer organisms are needed to cause death or disease.

The infectious dose

- The **infectious dose** of an organism required to cause disease varies greatly among the pathogenic bacteria.
- For example, *Shigella* and *Salmonella* both cause diarrhea by infecting the gastrointestinal tract, but the infectious dose of *Shigella* is less than 100 organisms, whereas the infectious dose of *Salmonella* is on the order of 100,000 organisms.

The infectious dose and virulence factors

- The infectious dose of bacteria depends primarily on their **virulence factors** (e.g., whether their pili allow them to adhere well to mucous membranes, whether they produce exotoxins or endotoxins, whether they possess a capsule to protect them from phagocytosis, and whether they can survive various nonspecific host defenses such as acid in the stomach).

Portals of Entry of Some Common Pathogens

- Respiratory tract: *Mycobacterium tuberculosis*
- Gastrointestinal tract: *Shigella dysenteriae*
- Skin: *Clostridium tetani*
- Genital tract :*Neisseria gonorrhoeae*

Surface Virulence Factors Important for Bacterial Pathogenesis

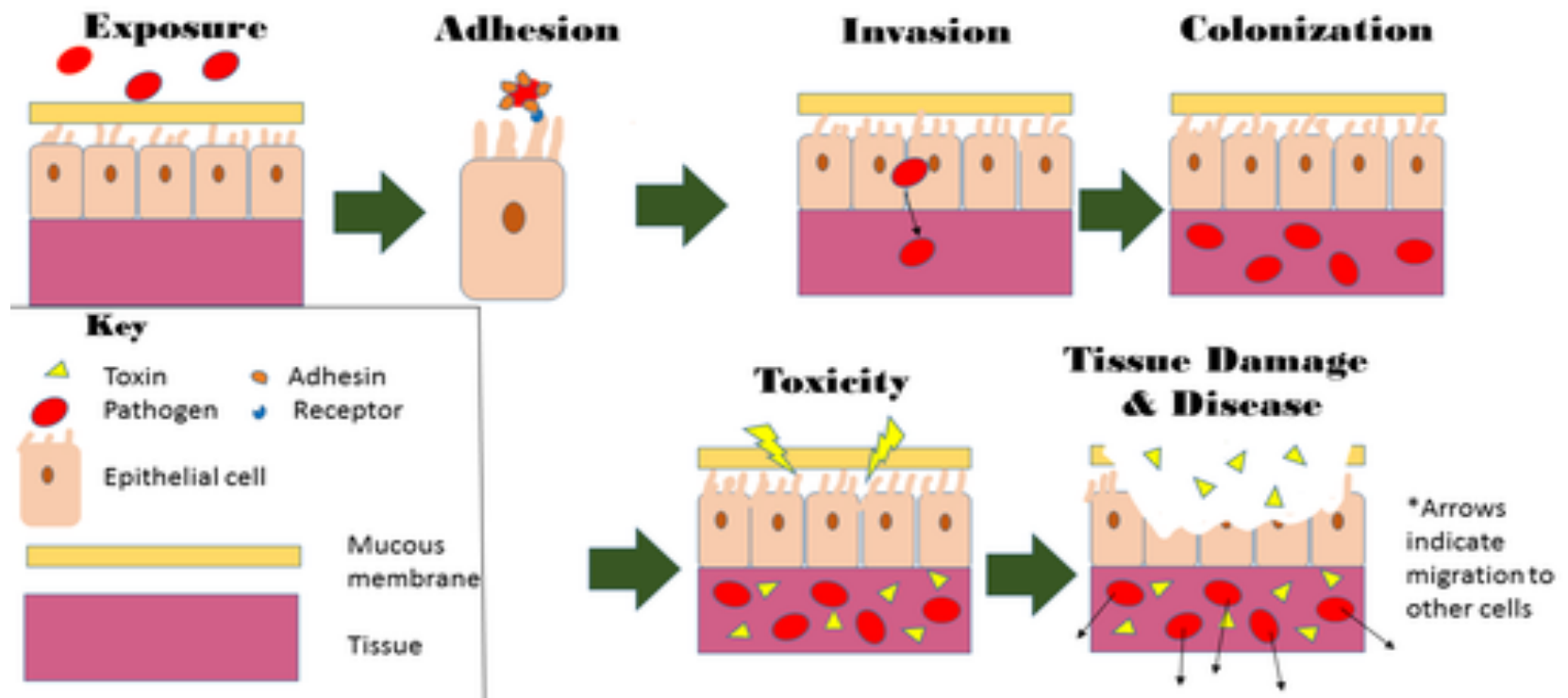
Organism	Virulence Factor
Gram-positive cocci <i>Streptococcus pneumoniae</i> <i>Streptococcus pyogenes</i> <i>Staphylococcus aureus</i>	Polysaccharide capsule M protein Protein A
Gram-negative cocci <i>Neisseria meningitides</i>	Polysaccharide capsule
Gram-positive rods <i>Bacillus anthracis</i>	Polypeptide capsule
Gram-negative rods <i>Haemophilus influenzae</i> <i>Klebsiella pneumoniae</i> <i>Escherichia coli</i> <i>Salmonella typhi</i> <i>Yersinia pestis</i>	Polysaccharide capsule Polysaccharide capsule Protein pili Polysaccharide capsule V and W proteins

Stages of Pathogenesis

- **Exposure/Initiation:** Contact with the bacteria.
- **Adhesion/Entry:** Bacteria attaches to and enters host tissues.
- **Invasion/Spread:** Local or systemic dissemination.
- **Colonization/ Multiplication:** Increase in the number of bacteria (for infectious diseases).
- **Evasion of Host Defenses:** How the bacteria avoids immunity.
- **Damage to Host:** The cause of symptoms (toxins, inflammation).
- **Outcome:** Recovery, chronic state, or death.

Stages of Pathogenesis

Pathogenic Infection



Adherence

- **Adherence (adhesion, attachment):** The process by which bacteria stick to the surfaces of host cells.
- After bacteria have entered the body, adherence is a major initial step in the infection process.
- The terms *adherence*, *adhesion*, and *attachment* are often used interchangeably.

Invasion: Microbiota: Nonpathogen:

- **Invasion:** The process whereby bacteria, animal parasites, fungi, and viruses enter host cells or tissues and spread in the body.
- **Microbiota:** Microbial flora harbored by normal, healthy individuals.
- **Nonpathogen:** A microorganism that does not cause disease; may be part of the normal microbiota.

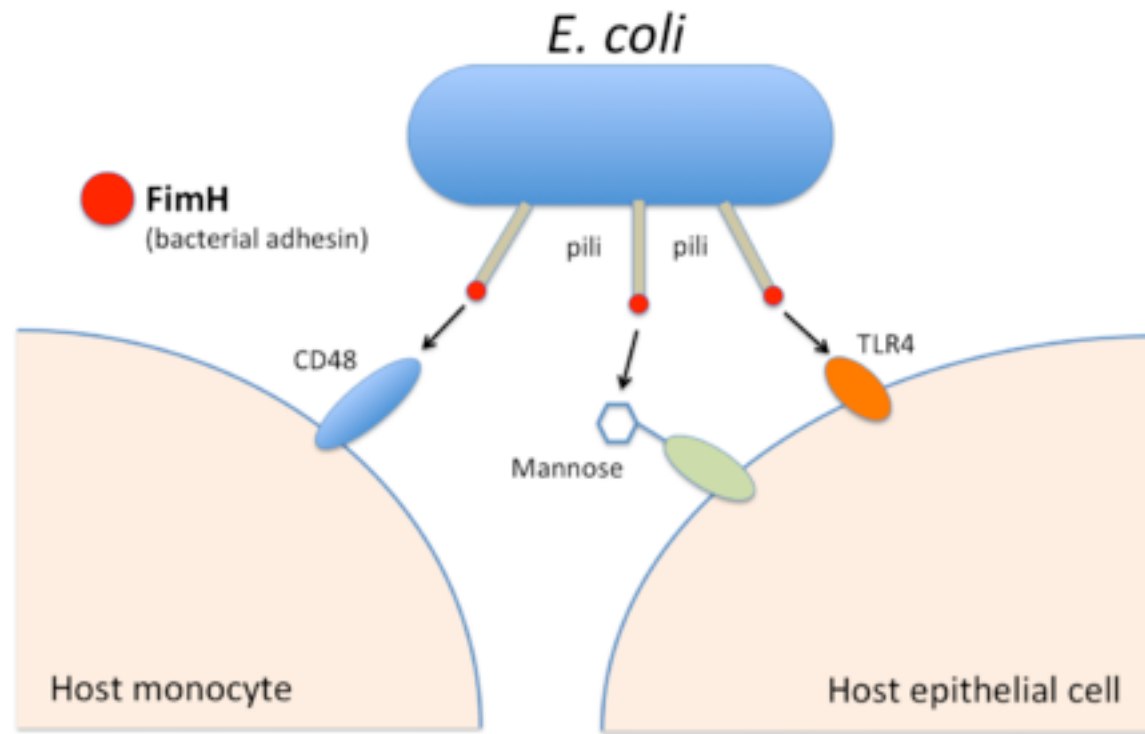
Colonization

Colonization: Important unless organism is traumatically implanted

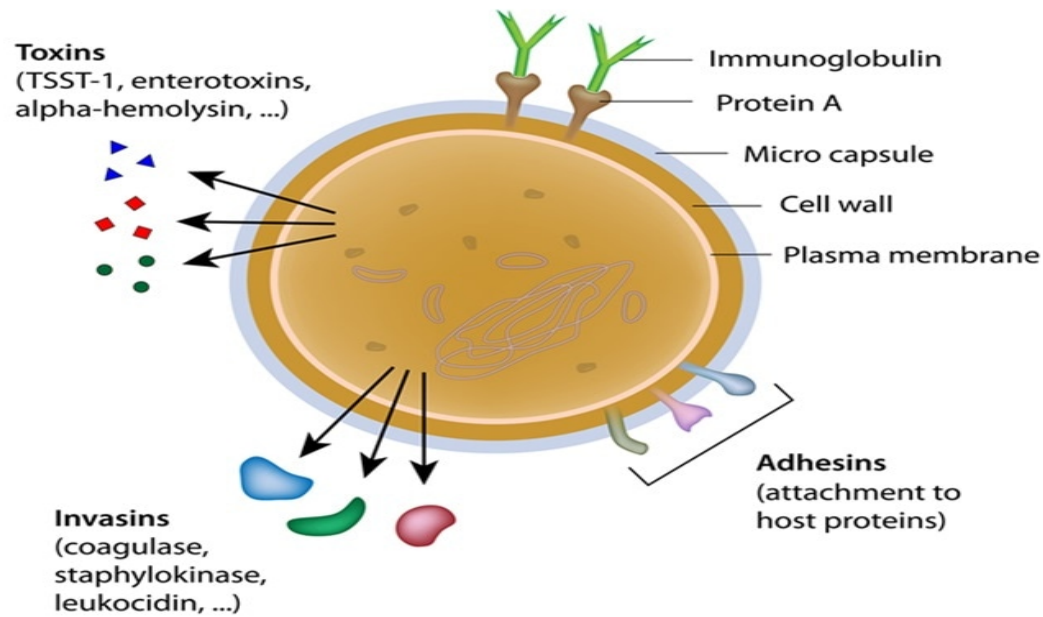
Adherence to cell surfaces involves:

- **Pili/fimbriae:** primary mechanism in most gram-negative cells.
- **Teichoic acids:** primary mechanism of gram-positive cells.
- **Adhesins:** colonizing factor adhesins, pertussis toxin, and hemagglutinins.
- **IgA proteases:** make it easier for pathogens to colonize.

FimH is a bacterial adhesin that helps bacteria such as *Escherichia coli* to bind to host cells and their receptors (here: the human proteins CD48 and TLR4, or mannose residues).



Staphylococcus aureus



Toxigenicity:

- **Toxigenicity:** The ability of a microorganism to produce a toxin that contributes to the development of disease.

TOXINS

Toxins may aid in

- invasiveness,
- damage cells,
- inhibit cellular processes,
- or trigger immune response and damage.

TOXINS

- Structural Toxins

Endotoxin (Lipopolysaccharide = LPS)

- – LPS is part of the **gram-negative outer membrane**.
- – **Toxic portion is lipid A**: generally not released (and toxic) until death of cell.
- .
- – **LPS is heat stable** and not strongly immunogenic so it **cannot be converted to a toxoid**.

Endotoxin

- Mechanism

- o **LPS activates macrophages**, leading to release of TNF-alpha, IL-1, and IL-6.
- o IL-1 is a major mediator of fever.
- o Macrophage activation and products lead to tissue damage.
- o Damage to the endothelium from **bradykinin-induced vasodilation leads to shock**.
- **Coagulation** (DIC) disseminated intravascular coagulation is mediated through the **activation of Hageman factor**
F.unction of Hageman factor is the initiation of the intrinsic pathway of blood clotting..

Exotoxins

Are **protein toxins**, generally **quite toxic** and **secreted by bacterial cells** (some gram +, some gram –)

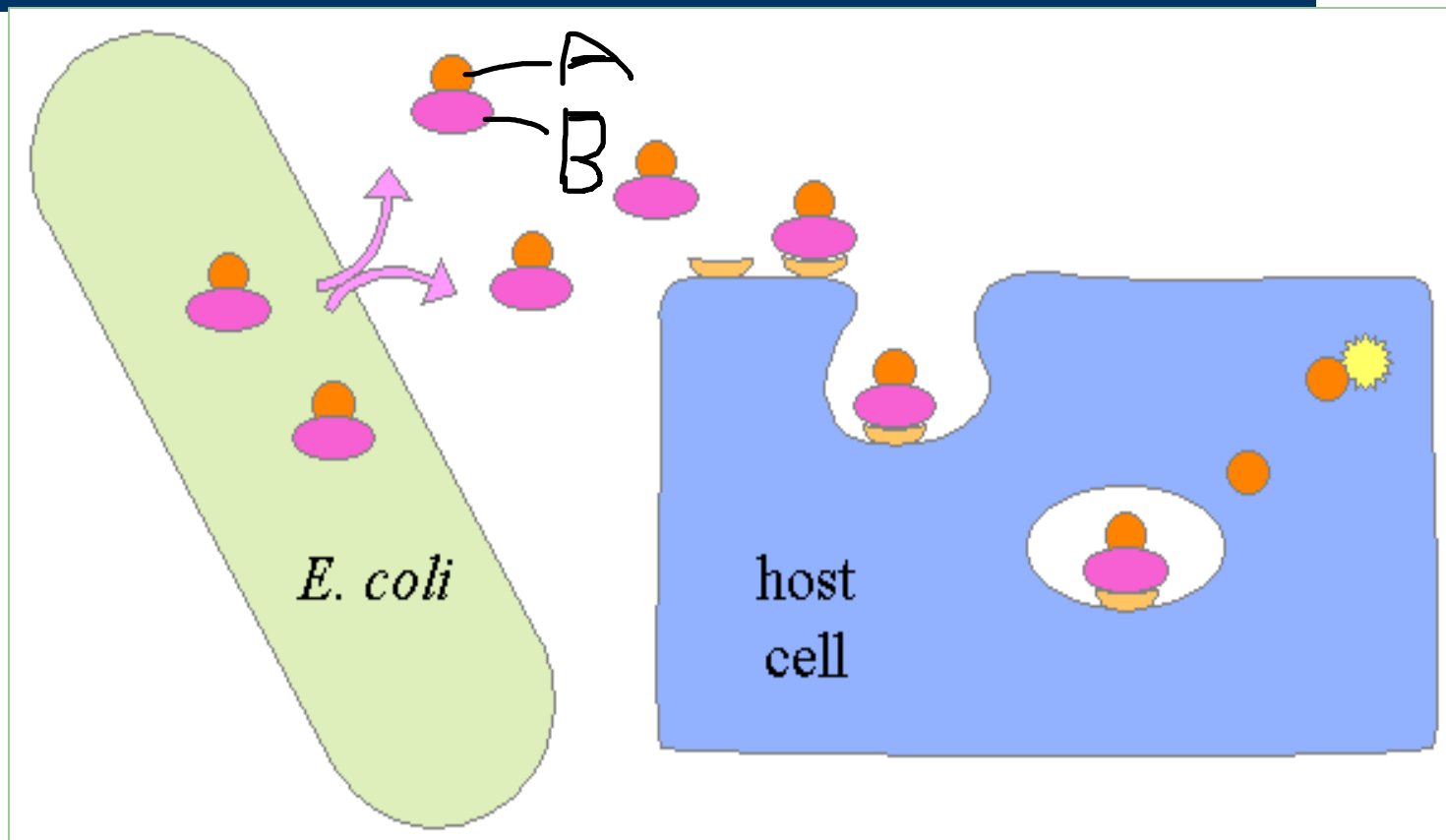
Can be modified by chemicals or heat to produce a **toxoid** that still is **immunogenic**, **but no longer toxic** so can be used as a vaccine

Exotoxins

A-B (or “two”) component protein toxins

- – **B** component **binds** to specific cell receptors to facilitate the internalization of A.
- – **A** component is the **active (toxic) component** (often an enzyme such as an ADP ribosyl transferase).

AB toxins



References

- 2019-Jawetz, Melnick & Adelberg's Medical Microbiology.
- Review of Medical Microbiology and Immunology 15th by Warren E. Levinson, Peter Chin-Hong, Elizabeth Joyce, Jesse Nussbaum, Brian Schwartz, 2018.
- Textbook of Diagnostic Microbiology 5th ed by Connie R. Mahon, Donald C. Lehman, George Manuselis, 2015